

A CONSTRUCTION OF 2-FILTERED BICOLIMITS OF CATEGORIES

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À la mémoire de Charles Ehresmann

Résumé. Nous définissons la notion de *2-catégorie 2-filtrante* et donnons une construction explicite du bicolimite d'un 2-foncteur à valeurs dans les catégories. Une catégorie considérée une 2-catégorie triviale est 2-filtrante si et seulement si c'est une catégorie filtrante, et notre construction conduit à une catégorie équivalente à la catégorie qui s'obtient par la construction usuelle de colimites filtrantes de catégories. Pour cette construction des axiomes plus faibles suffisent, et nous appelons la notion correspondante *2-catégorie pré 2-filtrante*. L'ensemble complet des axiomes est nécessaire pour montrer que les bicolimites 2-filtrantes ont les propriétés correspondantes aux propriétés essentielles des colimites filtrantes.

Introduction.

We define the notion of *2-filtered 2-category* and give an explicit construction of the bicolimit of a category valued 2-functor. A category considered as a trivial 2-category is 2-filtered if and only if it is a filtered category, and our construction yields a category equivalent to the category resulting from the usual construction of filtered colimits of categories. Weaker axioms suffice for this construction, and we call the corresponding notion *pre 2-filtered 2-category*. The full set of axioms is necessary to prove that 2-filtered bicolimits have the properties corresponding to the essential properties of filtered bicolimits.

In [3] Kennison already considers filterness conditions on a 2-category under the name of *bifiltered 2-category*. It is easy to check that a bifiltered 2-category is 2-filtered, so our results apply to bifiltered 2-categories. Actually Kennison's notion is equivalent to our's, but the other direction of this equivalence is not entirely trivial.

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1. Pre 2-Filtered 2-Categories and the construction LL

1.1 Definition. A 2-category $\mathcal{A}$ is defined to be *pre-2-filtered* when it satisfies the following two axioms:

F1.

$$\text{Given } \begin{array}{c} & A \\ f \nearrow & \\ E & \\ g \searrow & \\ & B \end{array} \quad \text{there exists invertible } \begin{array}{c} & A & \\ f \nearrow & & u \searrow \\ E & \gamma \Downarrow & C \\ g \searrow & & \nearrow v \\ & B & \end{array} .$$

F2.

$$\begin{array}{c} \text{Given any 2-cells} \end{array} \quad \begin{array}{c} & A & \\ f \nearrow & & u_1 \searrow \\ E & \gamma_1 \Downarrow & C_1 \\ g \searrow & & \nearrow v_1 \\ & B & \end{array}, \quad \begin{array}{c} & A & \\ f \nearrow & & u_2 \searrow \\ E & \gamma_2 \Downarrow & C_2 \\ g \searrow & & \nearrow v_2 \\ & B & \end{array}$$

$$\begin{array}{c} C_1 \\ \searrow w_1 \\ C \\ \nearrow w_2 \\ C_2 \end{array} \quad \text{there exists} \quad \begin{array}{c} C_1 \\ \searrow w_1 \\ C \\ \nearrow w_2 \\ C_2 \end{array} \quad \text{with invertible 2-cells } \alpha, \beta \text{ such that}$$

$$(1.2) \quad \begin{array}{c} & A & \\ f \nearrow & & u_1 \searrow \\ E & \gamma_1 \Downarrow & C_1 \\ g \searrow & & \nearrow v_1 \\ & B & \\ & \searrow v_2 & \nearrow w_2 \\ & & C_2 \end{array} \quad \alpha \Downarrow \quad \begin{array}{c} & A & \\ f \nearrow & & u_2 \searrow \\ E & \gamma_2 \Downarrow & C_2 \\ g \searrow & & \nearrow v_2 \\ & B & \\ & \searrow v_2 & \nearrow w_2 \\ & & C_2 \end{array} \quad \beta \Downarrow \quad \begin{array}{c} & C_1 & \\ u_1 \nearrow & & w_1 \searrow \\ & A & \\ f \nearrow & & u_2 \searrow \\ E & \gamma_2 \Downarrow & C_2 \\ g \searrow & & \nearrow v_2 \\ & B & \\ & \searrow v_2 & \nearrow w_2 \\ & & C_2 \end{array} \quad \gamma \Downarrow \quad \begin{array}{c} & C_1 & \\ u_1 \nearrow & & w_1 \searrow \\ & A & \\ f \nearrow & & u_2 \searrow \\ E & \gamma_2 \Downarrow & C_2 \\ g \searrow & & \nearrow v_2 \\ & B & \\ & \searrow v_2 & \nearrow w_2 \\ & & C_2 \end{array} .$$

1.3 Notation (the LL equation). Given two pairs of 2-cells (γ_1, α) , (γ_2, β) as in F2, we shall call the equation 1.2 the *equation LL*.

Axioms F1 and F2 can be weakened if we add a third axiom:

WF1. Same axiom F1 but not requiring γ to be invertible.

WF2. Same axiom F2 but only for invertible γ_1 and γ_2 .

WF3.

Given
$$\begin{array}{ccccc} & & A & & \\ f \nearrow & & & \searrow u & \\ E & & \gamma \Downarrow & & C \\ g \searrow & & & \nearrow v & \\ & & B & & \end{array}$$
 there exists invertible 2-cells α, β such that

$$\begin{array}{ccccc} & & A & & \\ f \nearrow & & & \searrow u & \\ E & & \gamma \Downarrow & & C \\ g \searrow & & & \nearrow v & \\ & & B & & \end{array} \begin{array}{c} \xrightarrow{h_2} \\ \alpha \Downarrow \\ \xrightarrow{h_1} \end{array} \begin{array}{ccccc} & & C_1 & & \\ s_1 \nearrow & & & \searrow w_1 & \\ A & & \beta \Downarrow & & D_1 \\ u \searrow & & & \nearrow h_1 & \\ C & & & & \\ g \nearrow & & & \searrow v & \\ & & B & & \end{array}$$
 and
$$\begin{array}{ccccc} & & C_1 & & \\ s_1 \nearrow & & & \searrow w_1 & \\ A & & \beta \Downarrow & & D_1 \\ u \searrow & & & \nearrow h_1 & \\ C & & & & \\ g \nearrow & & & \searrow v & \\ & & B & & \end{array}$$
 are invertible.

We leave to the reader the proof of the following:

1.4 Proposition. *The set of axioms F1, F2 is equivalent to the set WF1 WF2 and WF3. $\square$*

When $\mathcal{A}$ is a trivial 2-category (the only 2-cells are the identities), our axiom F1 corresponds to axiom PS1 in the definition of pseudofiltered category (cf [1] Exposé I), while axiom PS2 may not hold. Thus, a category which is pre 2-filtered as a trivial 2-category may not be pseudofiltered. Notice that our axioms F2 and WF3 are vacuous in this case.

Construction LL

Let $\mathcal{A}$ be a pre-2-filtered 2-category and $F : \mathcal{A} \rightarrow \mathcal{Cat}$ a category valued 2-functor. We shall now construct a category which is to be the bicolimit (in the sense made precise in theorem 1.19)

of F . This construction generalizes Grothendieck's construction of the category $\varinjlim_{\mathcal{A}} F$ for a filtered category $\mathcal{A}$ (cf [2], Exposé VI).

1.5 Definition (Quasicategory $\mathcal{L}(F)$).

i) An *object* is a pair (x, A) with $x \in FA$.

ii) A *premorphisms* $(x, A) \rightarrow (y, B)$ between two objects is a triple (u, ξ, v) , where $A \xrightarrow{u} C$, $B \xrightarrow{v} C$ and $\xi : F(u)(x) \rightarrow F(v)(y)$ in FC .

iii) A *homotopy* between two premorphisms is a quadruple $(w_1, w_2, \alpha, \beta) : (u_1, \xi_1, v_1) \Rightarrow (u_2, \xi_2, v_2)$, where $C_1 \xrightarrow{w_1} C$, $C_2 \xrightarrow{w_2} C$ and $\alpha : w_1 v_1 \xrightarrow{\cong} w_2 v_2$, $\beta : w_1 u_1 \xrightarrow{\cong} w_2 u_2$ are invertible 2-cells such that

$$\begin{array}{ccc} F(w_1)F(u_1)(x) = F(w_1 u_1)(x) & \xrightarrow{F(\beta)x} & F(w_2 u_2)(x) = F(w_2)F(u_2)(x) \\ \downarrow F(w_1)(\xi_1) & & \downarrow F(w_2)(\xi_2) \\ F(w_1)F(v_1)(y) = F(w_1 v_1)(y) & \xrightarrow{F(\alpha)y} & F(w_2 v_2)(y) = F(w_2)F(v_2)(y) \end{array}$$

commutes in FC .

We shall formally introduce now an *abuse of notation*

1.6 Notation.

i) We omit the letter F in denoting the action of F on its arguments.

Thus $A \xrightarrow[\alpha \Downarrow v]{u} B$ indicates a 2-cell in $\mathcal{A}$ as well as the correspond-

ing natural transformation $FA \xrightarrow[F(v)]{F(u)} FB$ in $\mathcal{Cat}$. In this way, the above commutative square becomes

$$\begin{array}{ccc} w_1 u_1 x & \xrightarrow{\beta x} & w_2 u_2 x \\ \downarrow w_1 \xi_1 & & \downarrow w_2 \xi_2 \\ w_1 v_1 y & \xrightarrow{\alpha y} & w_2 v_2 y \end{array}$$

ii) We write $F \xrightarrow{x} A$ in $(\mathcal{Cat}^{\mathcal{A}})^{op}$ for the natural transformation $\mathcal{A}[A, -] \rightarrow F$ defined by $x_C(A \xrightarrow{u} C) = F(u)(x) \in FC$.

Notice that the notation in i) $F(u)(x) = ux$ is consistent with this since juxtaposition denotes composition. Also, in the same vein, given an object $x \in FA$ and a functor $FA \xrightarrow{h} \mathcal{X}$ into any other category, the composite $F \xrightarrow{x} A \xrightarrow{h} \mathcal{X}$ makes sense and we have $h(F(x)) = hx$.

In this notation then, a premorphism $(x, A) \rightarrow (y, B)$ is a triple

(u, ξ, v) , where $F \begin{array}{ccc} & A & \\ x \nearrow & \xi \Downarrow & \searrow u \\ & B & \\ y \nearrow & & \searrow v \end{array} C$; that is $\xi : ux \longrightarrow vy$ in FC . A homotopy

between two premorphisms $F \begin{array}{ccc} & A & \\ x \nearrow & \xi_1 \Downarrow & \searrow u_1 \\ & B & \\ y \nearrow & & \searrow v_1 \end{array} C_1$, $F \begin{array}{ccc} & A & \\ x \nearrow & \xi_2 \Downarrow & \searrow u_2 \\ & B & \\ y \nearrow & & \searrow v_2 \end{array} C_2$ is a pair of in-

vertible 2-cells $B \begin{array}{ccc} & C_1 & \\ v_1 \nearrow & \alpha \Downarrow & \searrow w_1 \\ & C & \\ v_2 \nearrow & & \searrow w_2 \end{array} C_2$, $A \begin{array}{ccc} & C_1 & \\ u_1 \nearrow & \beta \Downarrow & \searrow w_1 \\ & C_2 & \\ u_2 \nearrow & & \searrow w_2 \end{array} C$ satisfying the LL equation:

$$F \begin{array}{ccccc} & & A & & \\ & x \nearrow & & u_1 \searrow & \\ & \xi_1 \Downarrow & & C_1 & \\ & & y \nearrow & & \\ & & B & & \\ & & v_1 \nearrow & & \\ & & \alpha \Downarrow & & \\ & & & C & \\ & & v_2 \nearrow & & \\ & & & C_2 & \end{array} = F \begin{array}{ccccc} & & & & C_1 \\ & & & u_1 \nearrow & \searrow w_1 \\ & & A & & \\ & x \nearrow & & u_2 \searrow & \\ & \xi_2 \Downarrow & & C_2 & \\ & & y \nearrow & & \\ & & B & & \\ & & v_2 \nearrow & & \\ & & & C & \end{array}.$$

We shall simply write $(\alpha, \beta) : \xi_1 \Rightarrow \xi_2$ for all the data involved in an homotopy.

At this point it is convenient to introduce the following notation:

1.7 Notation (LL-composition of 2-cells). Given three 2-cells α , β , and γ that fit into a diagram as it follows, we write:

$$\beta \circ_{\gamma} \alpha = \begin{array}{ccccc} & & \bullet & \searrow & \bullet \\ & \nearrow & \alpha \Downarrow & \nearrow & \bullet \\ \bullet & \xrightarrow{\quad} & \bullet & \searrow & \bullet \\ & \searrow & \beta \Downarrow & \nearrow & \bullet \\ & & \bullet & \nearrow & \bullet \end{array}$$

Thus, $\beta \circ_{\gamma} \alpha$ is our notation for the 2-cell between the top and the bottom composites of arrows. It should be thought of as the “composite of β with α over γ ”.

Homotopies compose: Consider a third premorphism $\begin{array}{ccc} & A & \\ x \nearrow & \xi_3 \Downarrow & \searrow u_3 \\ F & & C_3 \\ y \searrow & B & \nearrow v_3 \end{array}$

and an homotopy $\begin{array}{ccccc} & C_2 & & C_2 & \\ v_2 \nearrow & \alpha' \Downarrow & & \beta' \Downarrow & \searrow u_2 \\ B & & C' & & C' \\ v_3 \searrow & w'_3 \nearrow & & w'_3 \nearrow & \\ & C_3 & & C_3 & \end{array}$, $(\alpha', \beta') : \xi_2 \Rightarrow \xi_3$. Use

axiom F1 to obtain an invertible 2-cell $\begin{array}{ccc} & C & \\ w_2 \nearrow & \gamma \Downarrow & \searrow h \\ C_2 & & H \\ w'_2 \searrow & C' & \nearrow h' \end{array}$. This determines

a pair of 2-cells:

$$\begin{array}{ccccc} & C_1 & & C_1 & \\ v_1 \nearrow & \alpha \Downarrow & & \beta \Downarrow & \searrow w_1 \\ B & \xrightarrow{v_2} & C_2 & \xrightarrow{w_2} & C \\ v_3 \searrow & \alpha' \Downarrow & & \beta' \Downarrow & \searrow w_2 \\ & C_3 & & C_3 & \\ & w'_3 \nearrow & & w'_3 \nearrow & \\ & C' & & C' & \end{array} \xrightarrow{h} H \quad \text{and} \quad \begin{array}{ccccc} & C_1 & & C_1 & \\ u_1 \nearrow & \beta \Downarrow & & \alpha \Downarrow & \searrow w_1 \\ A & \xrightarrow{u_2} & C_2 & \xrightarrow{w_2} & C \\ u_3 \searrow & \beta' \Downarrow & & \alpha' \Downarrow & \searrow w_2 \\ & C_3 & & C_3 & \\ & w'_3 \nearrow & & w'_3 \nearrow & \\ & C' & & C' & \end{array} \xrightarrow{h} H$$

which defines a homotopy $\xi_1 \Rightarrow \xi_3$. The corresponding LL equation follows easily from the LL equations for (α, β) and (α', β') .

Using notation 1.7, we have:

1.8 Proposition (vertical composition of homotopies). *Given $(\alpha, \beta) : \xi_1 \Rightarrow \xi_2$ and $(\alpha', \beta') : \xi_2 \Rightarrow \xi_3$, there exists an appropriate γ and a homotopy $(\alpha' \circ_\gamma \alpha, \beta' \circ_\gamma \beta) : \xi_1 \Rightarrow \xi_3$.* $\square$

Homotopies are generated by composition out of two basic ones. The proof of the following is immediate:

1.9 Proposition. *Every pair of premorphisms ξ_1, ξ_2 and pair of 2-cells α, β that fit as follows determine two basic homotopies:*

$$\begin{array}{ccc}
 (\alpha, id_1) : F \begin{array}{c} \nearrow^x A \searrow^{u_1} \\ \xi_1 \Downarrow \\ \searrow^y B \nearrow^{v_1} \end{array} C_1 & \Rightarrow & F \begin{array}{c} \nearrow^x A \searrow^{u_1} \\ \xi_1 \Downarrow \\ \searrow^y B \nearrow^{v_1} \end{array} \begin{array}{c} C_1 \xrightarrow{w_1} \\ \alpha \Downarrow \\ C_2 \end{array} \\
 \\
 (id_2, \beta) : F \begin{array}{c} \nearrow^x A \searrow^{u_2} \\ \xi_2 \Downarrow \\ \searrow^y B \nearrow^{v_2} \end{array} C_2 & \Rightarrow & F \begin{array}{c} \nearrow^x A \searrow^{u_2} \\ \xi_2 \Downarrow \\ \searrow^y B \nearrow^{v_2} \end{array} C_2
 \end{array}$$

where id_1 and id_2 are the identity 2-cells corresponding to the arrows $w_1 u_1$ and $w_2 v_2$ respectively. When the pair (α, β) satisfies the LL equation, then the composite (over the identity 2-cell of the identity arrow of C_2) of these basic homotopies is defined, and it is equal to the homotopy determined by (α, β) . $\square$

Premorphisms compose: Given two premorphisms

$$(x, A) \xrightarrow{\xi} (y, B) \xrightarrow{\zeta} (z, C), \quad \begin{array}{ccc} & A & \\ x \nearrow & & \searrow u \\ F & \xi \Downarrow & S \\ & B & \nearrow v \\ & y & \end{array}, \quad \begin{array}{ccc} & B & \\ y \nearrow & & \searrow h \\ F & \zeta \Downarrow & T \\ & C & \nearrow k \\ & z & \end{array}, \quad \text{use}$$

axiom F1 to obtain invertible $\begin{array}{ccc} & S & \\ v \nearrow & & \searrow s \\ B & \gamma \Downarrow & H \\ & T & \nearrow t \\ & h & \end{array}$. According to 1.7 this

determines a premorphism $\zeta \circ_{\gamma} \xi$ between (x, A) and (z, C) that we take as a composite of ξ with ζ . Thus:

$$\zeta \circ_{\gamma} \xi = \begin{array}{ccccc} & & A & & \\ & x \nearrow & & \searrow u & \\ & & \xi \Downarrow & & \\ F & \xrightarrow{y} & B & \xrightarrow{h} & T \\ & \searrow z & \zeta \Downarrow & & \nearrow k \\ & & C & & \end{array} \begin{array}{ccc} & S & \\ & v \nearrow & \searrow s \\ & & \gamma \Downarrow \\ & & H \\ & h \nearrow & \searrow t \end{array}$$

We have:

1.10 Proposition (horizontal composition of premorphisms).
Given $\xi : (x, A) \rightarrow (y, B)$ and $\zeta : (y, B) \rightarrow (z, C)$, there exists an appropriate γ and $\zeta \circ_{\gamma} \xi : (x, A) \rightarrow (z, C)$. $\square$

Homotopies also compose horizontally:

1.11 Proposition (horizontal composition of homotopies).
Consider composable premorphisms and homotopies as follows:

$$(x, A) \xrightarrow{\xi_1} (y, B) \xrightarrow{\xi_2} (z, C) \quad \text{and} \quad (y, B) \xrightarrow{\zeta_1} (z, C) \xrightarrow{\zeta_2} (w, D)$$
Then, given any two composites

$$\zeta_1 \circ_{\gamma_1} \xi_1 \quad \text{and} \quad \zeta_2 \circ_{\gamma_2} \xi_2,$$
there exists an homotopy $\zeta_1 \circ_{\gamma_1} \xi_1 \Rightarrow \zeta_2 \circ_{\gamma_2} \xi_2$.

Proof. Consider the given homotopies and their LL equations:

$$\begin{array}{ccc}
 \begin{array}{ccccc}
 & & A & & \\
 & x \nearrow & & \searrow u_1 & \\
 F & & \xi_1 \Downarrow & S_1 & \\
 & y \searrow & & \nearrow v_1 & \\
 & & B & & \\
 & & & \searrow v_2 & \\
 & & & & S_2
 \end{array}
 & = &
 \begin{array}{ccccc}
 & & & & S_1 \\
 & & & u_1 \nearrow & \searrow s_1 \\
 & x \nearrow & A & & \\
 & & \xi_2 \Downarrow & S_2 & \\
 & y \searrow & & \nearrow v_2 & \\
 & & B & & \\
 & & & \searrow v_2 & \\
 & & & & S_2
 \end{array}
 \end{array}$$

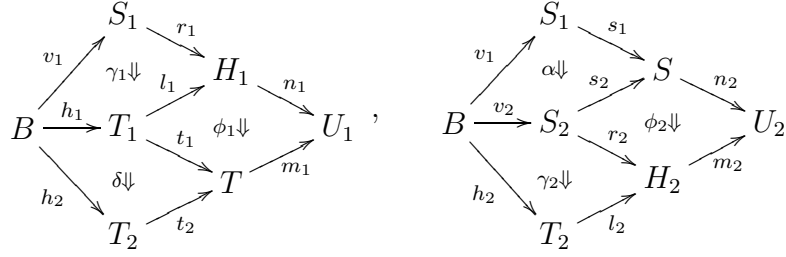
$$\begin{array}{ccc}
 \begin{array}{ccccc}
 & & B & & \\
 & y \nearrow & & \searrow h_1 & \\
 F & & \zeta_1 \Downarrow & T_1 & \\
 & z \searrow & & \nearrow k_1 & \\
 & & C & & \\
 & & & \searrow k_2 & \\
 & & & & T_2
 \end{array}
 & = &
 \begin{array}{ccccc}
 & & & & T_1 \\
 & & & h_1 \nearrow & \searrow t_1 \\
 & y \nearrow & B & & \\
 & & \zeta_2 \Downarrow & T_2 & \\
 & z \searrow & & \nearrow k_2 & \\
 & & C & & \\
 & & & \searrow k_2 & \\
 & & & & T_2
 \end{array}
 \end{array}$$

and consider the horizontal composites of the premorphisms:

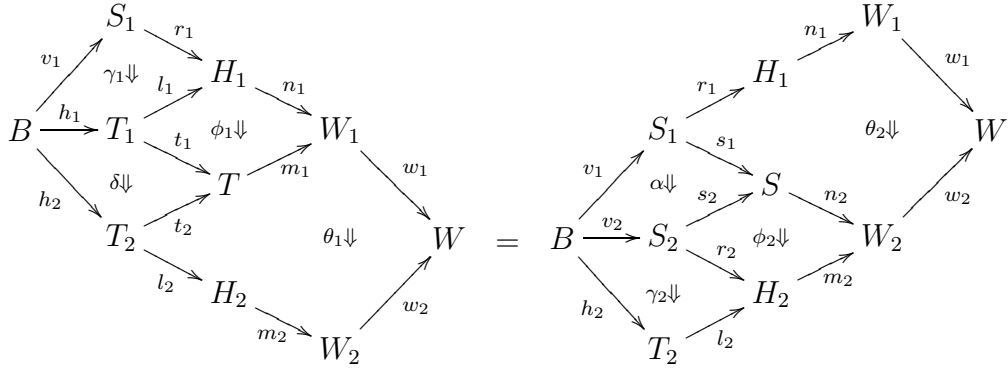
$$\begin{array}{ccc}
 \begin{array}{ccccccc}
 & & A & & & & \\
 & x \nearrow & & \searrow u_1 & & & \\
 F & & \xi_1 \Downarrow & S_1 & & r_1 & \\
 & y \searrow & & \nearrow v_1 & & \nearrow \gamma_1 \Downarrow & \\
 & & B & & h_1 & & H_1 \\
 & & & \searrow h_1 & & \nearrow l_1 & \\
 & & & & T_1 & & \\
 & & & & & \nearrow k_1 & \\
 & & & & C & &
 \end{array}
 & , &
 \begin{array}{ccccccc}
 & & A & & & & \\
 & x \nearrow & & \searrow u_2 & & & \\
 F & & \xi_2 \Downarrow & S_2 & & r_2 & \\
 & y \searrow & & \nearrow v_2 & & \nearrow \gamma_2 \Downarrow & \\
 & & B & & h_2 & & H_2 \\
 & & & \searrow h_2 & & \nearrow l_2 & \\
 & & & & T_2 & & \\
 & & & & & \nearrow k_2 & \\
 & & & & C & &
 \end{array}
 \end{array}$$

We shall produce a homotopy between these composites.

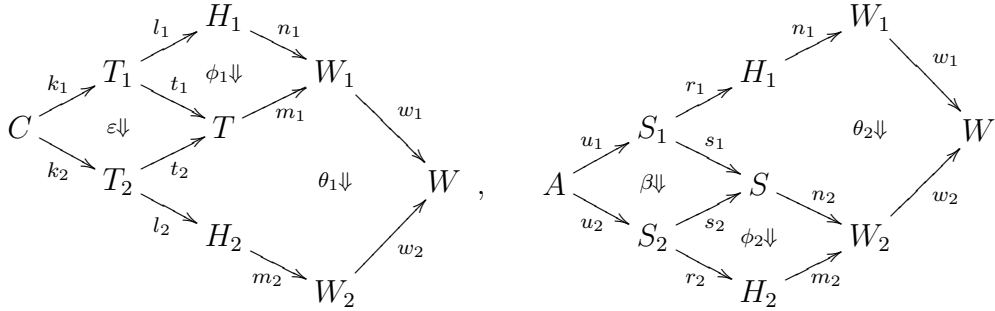
First use axiom F1 to obtain ϕ_1, ϕ_2 as follows:



Then use axiom F2 to obtain θ_1, θ_2 satisfying the LL equation:



The homotopy is given by the following pair of 2-cells:



The corresponding LL equation is :

$$\begin{array}{c}
 \begin{array}{ccccccc}
 & & A & \xrightarrow{u_1} & S_1 & \xrightarrow{r_1} & H_1 & \xrightarrow{n_1} & W_1 & \xrightarrow{w_1} & W \\
 & x \nearrow & \xi_1 \Downarrow & v_1 \nearrow & \gamma_1 \Downarrow & l_1 \nearrow & \phi_1 \Downarrow & m_1 \nearrow & \theta_1 \Downarrow & \\
 F & \xrightarrow{y} & B & & T_1 & & T & & W_1 & & W \\
 & z \searrow & \zeta_1 \Downarrow & h_1 \searrow & \varepsilon \Downarrow & t_2 \searrow & & & & & \\
 & & C & \xrightarrow{k_1} & T_2 & \xrightarrow{l_2} & H_2 & \xrightarrow{m_2} & W_2 & \xrightarrow{w_2} & W
 \end{array} \\
 = \\
 \begin{array}{ccccccc}
 & & A & \xrightarrow{u_1} & S_1 & \xrightarrow{r_1} & H_1 & \xrightarrow{n_1} & W_1 & \xrightarrow{w_1} & W \\
 & x \nearrow & \xi_2 \Downarrow & v_2 \nearrow & \beta \Downarrow & s_2 \nearrow & \phi_2 \Downarrow & m_2 \nearrow & \theta_2 \Downarrow & \\
 F & \xrightarrow{y} & B & & S_2 & & S & & W_1 & & W \\
 & z \searrow & \zeta_2 \Downarrow & h_2 \searrow & \gamma_2 \Downarrow & l_2 \searrow & & & & & \\
 & & C & \xrightarrow{k_2} & T_2 & \xrightarrow{l_2} & H_2 & \xrightarrow{m_2} & W_2 & \xrightarrow{w_2} & W
 \end{array}
 \end{array}$$

To pass from the left side to the right side of this equation use the LL equations of (ε, δ) , (θ_1, θ_2) and (α, β) , in this order. $\square$

1.12 Definition (equivalence of premorphisms). Two premorphisms ξ_1, ξ_2 are said to be *equivalent* when there exists a homotopy $(\alpha, \beta) : \xi_1 \Rightarrow \xi_2$. We shall write $\xi_1 \sim \xi_2$.

Equivalence is indeed an equivalence relation. Proposition 1.8 shows transitivity, the inverse 2-cells define an homotopy $(\alpha^{-1}, \beta^{-1}) : \xi_2 \Rightarrow \xi_1$ in the opposite direction, which shows symmetry, while reflexivity is obvious.

1.13 Definition (Category $\mathcal{L}(F)$). We define a category $\mathcal{L}(F)$ with objects pairs (x, A) , $x \in FA$. Morphisms are equivalence classes of premorphisms, and composition is defined by composing representative premorphisms.

It follows from 1.11 that composition is, up to equivalence, independent of the choice of representatives, and independent of the choice of the 2-cells given by axiom F1 when composing each pair of representatives. Since associativity holds and identities exist, this construction actually does define a category.

Some lemmas on pre-2-filtered categories

We establish now a few lemmas which are useful when proving the fundamental properties of the construction LL.

1.14 Lemma. *Given any pair of equivalent premorphisms*

$$\begin{array}{ccc} & A & \\ x \nearrow & & \searrow u_1 \\ F & \xrightarrow{\xi_1 \Downarrow} & C_1 \\ y \searrow & & \nearrow v_1 \\ & A & \end{array} \sim \begin{array}{ccc} & A & \\ x \nearrow & & \searrow u_2 \\ F & \xrightarrow{\xi_2 \Downarrow} & C_2 \\ y \searrow & & \nearrow v_2 \\ & A & \end{array} , \text{ if } u_1 = v_1 \text{ and } u_2 = v_2 , \text{ then}$$

we can choose an homotopy $(\alpha, \beta) : \xi_1 \Rightarrow \xi_2$ with $\alpha = \beta$.

Proof. It follows immediately from axiom F2. □

1.15 Lemma. *Given a finite family of 2-cells*

$$\begin{array}{ccc} & A & \\ f \nearrow & & \searrow u_i \\ E & \xrightarrow{\gamma_i \Downarrow} & C_i \\ g \searrow & & \nearrow v_i \\ & B & \end{array}$$

$i = 1 \dots n$, there exists $A \xrightarrow{u} C$, $B \xrightarrow{v} C$, $C_i \xrightarrow{w_i} C$, $i = 1 \dots n$, with invertible 2-cells α_i, β_i , such that the 2-cells

$$\begin{array}{ccccc} & A & & & \\ f \nearrow & & \searrow u & & \\ E & & & & C \\ \gamma_i \Downarrow & & \beta_i \Downarrow & & \\ g \searrow & & & & \\ & B & & & \\ & \nearrow v & & & \end{array} \quad \begin{array}{c} C_i \xrightarrow{w_i} C \\ \alpha_i \Downarrow \\ C_i \xrightarrow{w_i} C \end{array} \quad i = 1 \dots n$$

are all equal.

Given a second family of 2-cells
$$\begin{array}{ccc} & A & \\ h \nearrow & & \searrow u_i \\ H & \delta_i \Downarrow & C_i \\ l \searrow & & \nearrow v_i \\ & B & \end{array} \quad (\text{with same}$$

u_i, v_i, C_i), we can assume that the same $u, v, w_i, \alpha_i, \beta_i$, also equalize the 2-cells of the second family.

Proof. Axiom F2 provides the case $n = 2$ with $u = w_1 u_1$, $v = w_2 v_2$, $\alpha_1 = \alpha$, $\beta_1 = id$, $\alpha_2 = id$, and $\beta_2 = \beta$. Using this case, induction is straightforward. For the second part, if the 2-cells of the second family are not yet equalized, use the lemma again (and patch the new 2-cells also into the first family) $\square$

From axiom F2 we deduce

1.16 Lemma. Given any 2-cells
$$\begin{array}{ccc} & A & \\ f \nearrow & & \searrow u_1 \\ E & \gamma_1 \Downarrow & C_1 \\ g \searrow & & \nearrow v_1 \\ & B & \end{array} \quad \begin{array}{ccc} & A & \\ f \nearrow & & \searrow u_2 \\ E & \gamma_2 \Downarrow & C_2 \\ g \searrow & & \nearrow v_2 \\ & B & \end{array}$$
 and an object $F \xrightarrow{x} E$, the premorphisms
$$\begin{array}{ccc} & A & \\ fx \nearrow & & \searrow u_1 \\ F & \gamma_1 x \Downarrow & C_1 \\ gx \searrow & & \nearrow v_1 \\ & B & \end{array} \quad \begin{array}{ccc} & A & \\ fx \nearrow & & \searrow u_2 \\ F & \gamma_2 x \Downarrow & C_2 \\ gx \searrow & & \nearrow v_2 \\ & B & \end{array}$$
 are equivalent. $\square$

From proposition 1.9 it follows that

1.17 Lemma. Given a pair of premorphism ξ_1, ξ_2 and a pair of invertible two cells α, β that fit as follows, we have:

$$\begin{array}{ccc} & A & \\ x \nearrow & & \searrow u_1 \\ F & \xi_1 \Downarrow & C_1 \\ y \searrow & & \nearrow v_1 \\ & B & \end{array} \quad \sim \quad \begin{array}{ccccc} & A & & & \\ x \nearrow & & \searrow u_1 & & \\ F & \xi_1 \Downarrow & C_1 & \xrightarrow{w_1} & \\ y \searrow & & \nearrow v_1 & & \\ & B & & \searrow \alpha \Downarrow & \\ & & v_2 & \nearrow w_2 & \\ & & & & C_2 \end{array} \quad \text{and}$$

$$\begin{array}{ccc}
& C_1 & \\
u_1 \nearrow & & \searrow w_1 \\
& A & \xrightarrow{\beta \Downarrow} C \\
x \nearrow & & \searrow u_2 \\
F & \xrightarrow{\xi_2 \Downarrow} C_2 & \\
y \searrow & & \nearrow w_2 \\
& B &
\end{array}
\sim
\begin{array}{ccc}
& A & \\
x \nearrow & & \searrow u_2 \\
F & \xrightarrow{\xi_2 \Downarrow} C_2 & \\
y \searrow & & \nearrow v_2 \\
& B &
\end{array}$$

When the pair α, β satisfy the LL equation, then transitivity applied to these two equivalences yields the equivalence $\xi_1 \sim \xi_2$. $\square$

From this lemma and transitivity of equivalence we deduce

1.18 Lemma. *Given a premorphism ξ , and a pair of invertible two cells α, β that fit as follows, we have:*

$$\begin{array}{ccc}
& A & \\
x \nearrow & & \searrow u \\
F & \xrightarrow{\xi \Downarrow} C & \\
y \searrow & & \nearrow v \\
& B &
\end{array}
\sim
\begin{array}{ccccc}
& A & & & \\
x \nearrow & & \searrow u & \xrightarrow{s} & D \\
F & \xrightarrow{\xi \Downarrow} C & \xrightarrow{w} & & \\
y \searrow & & \nearrow v & \xrightarrow{\beta \Downarrow} & D \\
& B & & \nearrow t &
\end{array}$$

$\square$

The universal property of the construction LL

A *pseudococone* for a 2-functor F with vertex the category $\mathcal{X}$ is a pseudonatural transformation $F \xRightarrow{h} \mathcal{X}$ from F to the 2-functor which is constant at $\mathcal{X}$. Explicitly, it consists of a family of functors $(h_A : FA \rightarrow \mathcal{X})_{A \in \mathcal{A}}$, and a family of invertible natural transformations $(h_u : h_B \circ u \rightarrow h_A)_{(A \xrightarrow{u} B) \in \mathcal{A}}$. A morphism $h \xRightarrow{\varphi} l$ of pseudococones (with same vertex) is a modification; as such, it consists of a family of natural transformations $(h_A \xRightarrow{\varphi_A} l_A)_{A \in \mathcal{A}}$. In accordance with notation 1.6, we have

$$\begin{array}{ccc}
A & & \\
u \downarrow & \searrow h_A & \\
& & \mathcal{X} \\
B & \xrightarrow{h_B} &
\end{array}
\quad
\begin{array}{ccc}
A & \xrightarrow{h_A} & \mathcal{X} \\
& \varphi_A \Downarrow & \\
& l_A &
\end{array}$$

This data is subject to the equations:

PC0.
PC1.

$$h_{id_A} = id_{h_A}$$

$$\begin{array}{ccc} A & \searrow h_A & \\ u \downarrow & h_u \uparrow & \\ B & \xrightarrow{h_B} & \mathcal{X} \\ v \downarrow & h_v \uparrow & \\ C & \nearrow h_C & \end{array} = \begin{array}{ccc} A & \searrow h_A & \\ u \downarrow & h_{vu} \uparrow & \\ B & \xrightarrow{h_B} & \mathcal{X} \\ v \downarrow & & \\ C & \nearrow h_C & \end{array}$$

PC2.

$$\begin{array}{ccc} A & \searrow h_A & \\ u \downarrow \gamma \Rightarrow v \downarrow & h_v \uparrow & \\ B & \xrightarrow{h_B} & \mathcal{X} \end{array} = \begin{array}{ccc} A & \searrow h_A & \\ u \downarrow & h_u \uparrow & \\ B & \xrightarrow{h_B} & \mathcal{X} \end{array}$$

PCM.

$$\begin{array}{ccc} A & \searrow l_A & \\ u \downarrow & h_A \uparrow & \\ B & \xrightarrow{h_B} & \mathcal{X} \end{array} = \begin{array}{ccc} A & \searrow l_A & \\ u \downarrow & l_u \uparrow & \\ B & \xrightarrow{\varphi_B \uparrow} & \mathcal{X} \\ & h_B & \end{array}$$

Given a pseudo cone $F \xRightarrow{h} \mathcal{Z}$ and a 2-functor $\mathcal{Z} \xrightarrow{s} \mathcal{X}$, it is clear and straightforward how to define a pseudocone $F \xRightarrow{sh} \mathcal{X}$ which is the composite $F \xRightarrow{h} \mathcal{Z} \xrightarrow{s} \mathcal{X}$; this determines a functor $Cat[\mathcal{Z}, \mathcal{X}] \times \mathcal{PC}[F, \mathcal{Z}] \longrightarrow \mathcal{PC}[F, \mathcal{X}]$.

1.19 Theorem. Let $A \xrightarrow{u} B$ in FA and $x \xrightarrow{\xi} y$ in $\mathcal{A}$. The following formulas define a pseudocone $F \xRightarrow{\lambda} \mathcal{L}(F)$:

$$\lambda_A(x) = F \xrightarrow{x} A, \quad \lambda_A(\xi) = F \begin{array}{ccc} & A & \\ x \nearrow & & \searrow id \\ & A & \\ y \searrow & & \nearrow id \end{array} \xi \Downarrow A, \quad \lambda_u(x) = F \begin{array}{ccc} & B & \\ ux \nearrow & & \searrow id \\ & B & \\ x \searrow & & \nearrow u \end{array} id \Downarrow B$$

which induces by composition an equivalence of categories $Cat[\mathcal{L}(F), \mathcal{X}] \xrightarrow{\cong} \mathcal{PC}[F, \mathcal{X}]$, and this equivalence is actually an isomorphism. We have:

for all h there exists a unique $\tilde{h}$ such that $\tilde{h} \lambda = h$: $F \xRightarrow{\lambda} \mathcal{L}(F)$.

$$\begin{array}{ccc} F & \xRightarrow{\lambda} & \mathcal{L}(F) \\ & \searrow \forall h & \downarrow \exists ! \tilde{h} \\ & & \mathcal{X} \end{array}$$

Proof. Functoriality of λ_A , naturality of λ_u and equation PC1 hold tautologically. The validity of PC2 means the following equivalence of premorphisms:

$$\begin{array}{ccccc} & & B & \xrightarrow{id} & \\ & \nearrow ux & \downarrow \gamma x & \searrow id & \\ F & \xrightarrow{vx} & B & \xrightarrow{id} & B \\ & \searrow x & \downarrow id & \nearrow id & \\ & & A & \xrightarrow{v} & B \end{array} \quad \sim \quad \begin{array}{ccccc} & & B & \xrightarrow{id} & \\ & \nearrow ux & \downarrow id & \searrow u & \\ F & \xrightarrow{x} & A & \xrightarrow{u} & B \end{array}$$

which is given by lemma 1.16.

We pass now to prove the universal property. Given $F \xRightarrow{h} \mathcal{X}$,

define $\tilde{h}$ by the formulas: for $F \xrightarrow{x} A \xrightarrow{u} C$ in $\mathcal{L}(F)$,

$$\tilde{h}(x) = (F \xrightarrow{x} A \xrightarrow{h_A} \mathcal{X}), \quad \tilde{h}(\xi) = F \begin{array}{ccc} \xrightarrow{x} A & \xrightarrow{u} C & \xrightarrow{h_C} \mathcal{X} \\ \downarrow \xi & \downarrow h_u^{-1} & \\ y \rightarrow B & \xrightarrow{v} C & \downarrow h_v \\ & & \mathcal{X} \end{array} \begin{array}{c} \xrightarrow{h_A} \\ \xrightarrow{h_B} \end{array}$$

We have to show that the definition of $\tilde{h}(\xi)$ is compatible with the equivalence of premorphisms. It is enough to consider the two cases in

lemma 1.17. For the first case we have to show the equation

$$(1) \quad \begin{array}{c} \begin{array}{ccccc} & & A & & \\ & x \nearrow & & \searrow u & \\ F & & \xi \Downarrow & C & \\ & y \searrow & & \nearrow v & \\ & & B & & \\ & & & & \searrow h_C \\ & & & & \mathcal{X} \end{array} \end{array} \quad = \quad \begin{array}{c} \begin{array}{ccccc} & & A & & \\ & x \nearrow & & \searrow u & \\ F & & \xi \Downarrow & C & \\ & y \searrow & & \nearrow v & \\ & & B & & \\ & & & & \searrow h_C \\ & & & & \mathcal{X} \end{array} \end{array}$$

Consider the following equation which follows from PC1:

$$(2) \quad \begin{array}{c} \begin{array}{ccccc} & & A & & \\ & x \nearrow & & \searrow u & \\ F & & \xi \Downarrow & C & \\ & y \searrow & & \nearrow v & \\ & & B & & \\ & & & & \searrow h_C \\ & & & & \mathcal{X} \end{array} \end{array} \quad = \quad \begin{array}{c} \begin{array}{ccccc} & & A & & \\ & x \nearrow & & \searrow u & \\ F & & \xi \Downarrow & C & \\ & y \searrow & & \nearrow v & \\ & & B & & \\ & & & & \searrow h_C \\ & & & & \mathcal{X} \end{array} \end{array}$$

The right hand sides of equations (1) and (2) are equal by PC2, while the left hand sides are clearly equal. The second case in lemma 1.17 is treated in a similar manner. Functoriality of $\tilde{h}(\xi)$ follows from PC1 and PC2 with the same type of techniques as above. Finally, the equation $\tilde{h} \lambda = h$ is immediate for the whole cone structure. $\square$

We finish this section with a lemma which follows from lemma 1.14

1.20 Lemma. *Given two arrows $x \xrightarrow{\xi_1} y$ in FA , if $\lambda_A(x) = \lambda_A(y)$ in $\mathcal{L}(F)$, then there exists $A \xrightarrow{w} C$ such that $wx = wy$ in FC . $\square$*

2. 2-Filtered 2-Categories

2.1 Definition. A 2-category $\mathcal{A}$ is defined to be *pseudo 2-filtered* when it is pre 2-filtered and satisfies the stronger form of axiom F1:

FF1.

$$\text{Given } \begin{array}{c} f_1 \nearrow A \\ E_1 \\ g_1 \searrow B \end{array}, \begin{array}{c} f_2 \nearrow A \\ E_2 \\ g_2 \searrow B \end{array} \text{ there exists } \begin{array}{c} f_1 \nearrow A \searrow u \\ E_1 \xrightarrow{\gamma_1 \Downarrow} C \\ g_1 \searrow B \nearrow v \end{array}, \begin{array}{c} f_2 \nearrow A \searrow u \\ E_2 \xrightarrow{\gamma_2 \Downarrow} C \\ g_2 \searrow B \nearrow v \end{array},$$

with γ_1 and γ_2 invertible 2-cells.

It is defined to be *2-filtered* when it is pseudo 2-filtered, non empty, and satisfies in addition the following axiom.

F0.

$$\text{Given } \begin{array}{c} A \\ \\ B \end{array} \text{ there exists } \begin{array}{c} A \searrow u \\ \\ B \nearrow v \\ C \end{array}.$$

As was the case for axiom F1, in the presence of axiom WF3, axiom FF1 can be replaced by the weaker version in which we do not require the 2-cells γ_1 and γ_2 to be invertible.

When $\mathcal{A}$ is a trivial 2-category (the only 2-cells are the identities), axiom F0 is the usual axiom in the definition of filtered category, while our axiom FF1 is equivalent to the conjunction of the two axioms PS1 and PS2 in the definition of pseudofiltered category (cf [1] Exposé I).

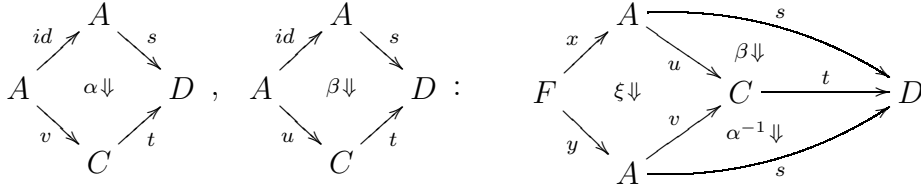
Two properties of the construction LL that follow for pseudo 2-filtered 2-categories and not for pre 2-filtered 2-categories are the following:

2.2 Lemma. *Given any morphism $(x, A) \rightarrow (y, A)$ in $\mathcal{L}(F)$, we can*

choose a representative premorphism $F \xrightarrow{\xi \Downarrow} C$ with $u = v$.

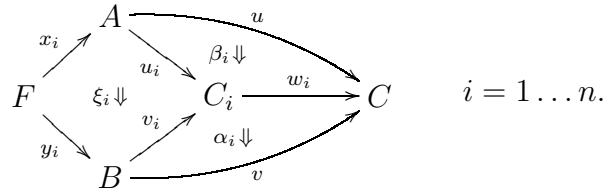
Proof. Consider $\begin{array}{c} x \nearrow A \searrow u \\ F \xrightarrow{\xi \Downarrow} C \\ y \searrow A \nearrow v \end{array}$ and apply axiom FF1 to obtain invertible

2-cells α, β as follows:



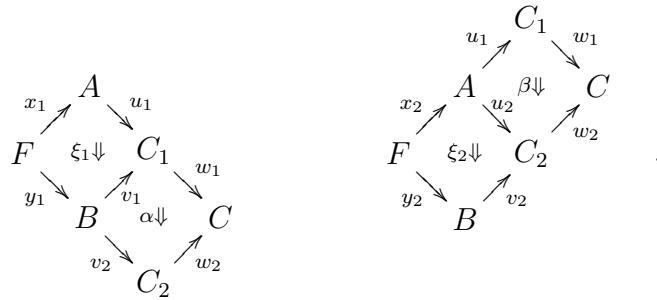
The proof follows by lemma 1.18. $\square$

2.3 Lemma. Given a finite family of premorphisms $F \xrightarrow{x_i} A \xrightarrow{u_i} C_i$, $F \xrightarrow{y_i} B \xrightarrow{v_i} C_i$, $i = 1 \dots n$, there exists $A \xrightarrow{u} C$, $B \xrightarrow{v} C$, $C_i \xrightarrow{w_i} C$, $i = 1 \dots n$, with invertible 2-cells α_i, β_i as in the diagram:



When $A = B$ we can assume $u = v$.

Proof. Given ξ_1, ξ_2 , apply FF1 to obtain invertible α, β to fit as follows:



This gives the case $n = 2$ with $u = w_1 u_1$, $v = w_2 v_2$, $\alpha_1 = \alpha$, $\beta_1 = id$, $\alpha_2 = id$, and $\beta_2 = \beta$. Using this case, induction is straightforward. For the second part, do as in the proof of lemma 2.2. $\square$

2.4 Theorem. Let $\mathcal{A}$ be a pre 2-filtered 2-category, $F : \mathcal{A} \longrightarrow \mathcal{Cat}$ a 2-functor, and $\mathcal{P}$ a finite category. Consider the 2-functor $F^{\mathcal{P}} : \mathcal{A} \longrightarrow \mathcal{Cat}$ defined by $F^{\mathcal{P}}(A) = (FA)^{\mathcal{P}}$, and the canonical functor:

$$\diamond : \mathcal{L}(F^{\mathcal{P}}) \longrightarrow \mathcal{L}(F)^{\mathcal{P}} \quad (\text{given by theorem 1.19}).$$

Then, $\diamond$ is an equivalence of categories provided that $\mathcal{A}$ is 2-filtered or that $\mathcal{A}$ is pseudo 2-filtered and $\mathcal{P}$ is connected.

Proof. Notice that an object $F^{\mathcal{P}} \longrightarrow A$ in $\mathcal{L}(F^{\mathcal{P}})$ is by definition a diagram $\mathcal{P} \longrightarrow FA$. We shall prove, in turn, that $\diamond$ is (a) essentially surjective, (b) faithful, and (c) full.

(a) *essentially surjective:* We shall see that given a diagram $\mathcal{P} \rightarrow \mathcal{L}(F)$, there exists $A \in \mathcal{A}$ and a factorization (up to isomorphism):

$$\begin{array}{ccc} & \mathcal{P} & \\ \swarrow & & \downarrow \\ FA & \xrightarrow{\lambda_A} & \mathcal{L}(F) \end{array}$$

Consider explicitly an object in $\mathcal{L}(F)^{\mathcal{P}}$:

$$F \xrightarrow{x_k} A_k, \quad k \in \mathcal{P}, \quad \begin{array}{ccc} & A_p & \\ x_p \nearrow & & \searrow u_f \\ F & \varphi_f \Downarrow & A_f \\ x_q \searrow & & \nearrow v_f \\ & A_q & \end{array}, \quad p \xrightarrow{f} q \in \mathcal{P}.$$

satisfying equations $\varphi_{f \circ g} \sim \varphi_f \circ_{\gamma} \varphi_g$ for all composable pairs f, g .

Let $\mathcal{Q} \subset \mathcal{P}$ be a part of $\mathcal{P}$ for which there exists A, w_k, ψ_f such that:

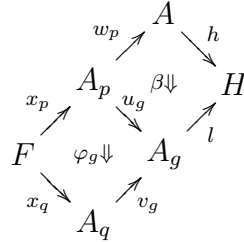
$$F \xrightarrow{x_k} A_k, \quad k \in \mathcal{Q}, \quad \begin{array}{ccc} & A_p & \\ x_p \nearrow & & \searrow u_f \\ F & \varphi_f \Downarrow & A_f \\ x_q \searrow & & \nearrow v_f \\ & A_q & \end{array} \sim \begin{array}{ccc} & A_p & \\ x_p \nearrow & & \searrow w_p \\ F & \psi_f \Downarrow & A \\ x_q \searrow & & \nearrow w_q \\ & A_q & \end{array}, \quad p \xrightarrow{f} q \in \mathcal{Q}.$$

$\mathcal{Q}$ is not necessarily a subcategory, but we agree that if $p \xrightarrow{f} q \in \mathcal{Q}$, then we consider $p \in \mathcal{Q}$ and $q \in \mathcal{Q}$. The equations $\psi_{f \circ g} \sim \psi_f \circ_{\gamma} \psi_g$

hold for all composable pairs f, g in $\mathcal{Q}$ with $f \circ g$ also in $\mathcal{Q}$. By lemma 1.20 we can assume strict equality $\psi_{f \circ g} = \psi_f \circ \psi_g$ in FA .

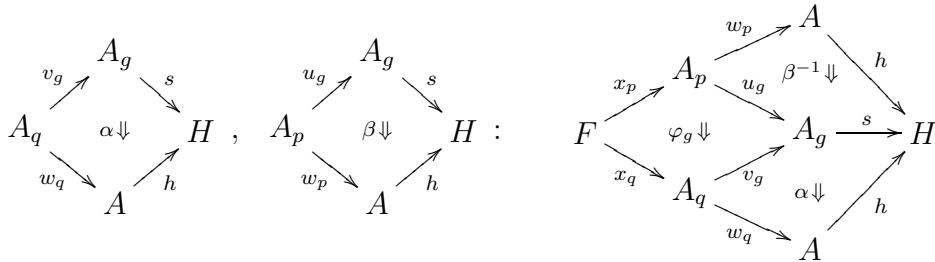
We shall see that if $p \xrightarrow{g} q$ is not in $\mathcal{Q}$, we can add it to $\mathcal{Q}$ in such a way that the enlarged part retains the same property. In what it follows we use without mention lemma 1.18.

1) $p \in \mathcal{Q}, q \notin \mathcal{Q}$ or $q \in \mathcal{Q}, p \notin \mathcal{Q}$: In the first case apply axiom F1 to obtain an invertible 2-cell β as follows:



The new A is H , the new w_k are hw_k , all $k \in \mathcal{Q}$, the new ψ_f are $h\psi_f$, all $f \in \mathcal{Q}$, and, finally, $w_q = lv_g$, and ψ_g is the 2-cell above. The other case can be proved in the same way.

2) $p \in \mathcal{Q}, q \in \mathcal{Q}$: Apply axiom FF1 to obtain invertible 2-cells α, β as follows:



The new A is H , the new w_k are hw_k , all $k \in \mathcal{Q}$, the new ψ_f are $h\psi_f$, all $f \in \mathcal{Q}$, and, finally, ψ_g is the 2-cell above on the right. This proof holds whether $p \neq q$ or $p = q$.

3) $p \notin \mathcal{Q}, q \notin \mathcal{Q}$: Apply axiom F0 to obtain $A_g \xrightarrow{w} H, A \xrightarrow{h} H$. The new A is H , the new w_k are hw_k , all $k \in \mathcal{Q}$, the new ψ_f are $h\psi_f$, all $f \in \mathcal{Q}$, and, finally, $w_p = lu_g, w_q = lv_g, \psi_g = l\varphi_g$. If $p = q$, use lemma 2.2 to assume $u_g = v_g$, and in this way w_p is uniquely defined.

Notice that if $\mathcal{P}$ is connected, it is not necessary to consider this case since we can always choose $p \xrightarrow{g} q$ such that either p , or q , or both, are in $\mathcal{Q}$. Thus for connected $\mathcal{P}$ axiom F0 is not necessary.

It is clear that any singleton $\{(k, f = id_k)\}$ serves as an initial $\mathcal{Q}$, thus we can assume $\mathcal{Q} = \mathcal{P}$.

To finish the proof observe that given any arrow $ux \xrightarrow{\psi} vy$ in FA , the square $(ux, A) \xrightarrow[\cong]{(id, id, u)} (x, A)$ commutes in $\mathcal{L}(F)$.

$$\begin{array}{ccc} (ux, A) & \xrightarrow[\cong]{(id, id, u)} & (x, A) \\ \downarrow \lambda_A(\psi) & & \downarrow (u, \psi, v) \\ (vy, A) & \xrightarrow[\cong]{(id, id, v)} & (y, B) \end{array}$$

Notice that if $\mathcal{P}$ is a poset, case 2) cannot happen, so axiom FF1 is not necessary. For posets $\mathcal{P}$ the functor $\diamond$ is essentially surjective also for pre 2-filtered 2-categories.

(b) *faithful*: Consider two premorphisms $F^{\mathcal{P}} \xrightarrow{\xi} C, F^{\mathcal{P}} \xrightarrow{\eta} D$

in $\mathcal{L}(F^{\mathcal{P}})$, $F \xrightarrow{\xi_k} C, F \xrightarrow{\eta_k} D, k \in \mathcal{P}$. To be equivalent in $\mathcal{L}(F)^{\mathcal{P}}$ means that there are homotopies $(\alpha_k, \beta_k) : \xi_k \Rightarrow \eta_k$ given by

invertible 2-cells $B \xrightarrow{v} C \xrightarrow{w_k} H_k, A \xrightarrow{u} C \xrightarrow{w_k} H_k, k \in \mathcal{P}$. From lemma

1.15 it readily follows that we can assume there are single invertible 2-cells α, β which define all the homotopies $(\alpha, \beta) : \xi_k \Rightarrow \eta_k$. But this means that ξ and η are equivalent in $\mathcal{L}(F^{\mathcal{P}})$.

Notice that this proof of the faithfulness of the functor $\diamond$ holds for pre 2-filtered 2-categories.

(c) *full*: Consider two objects $F^{\mathcal{P}} \xrightarrow{x} A, F^{\mathcal{P}} \xrightarrow{y} B$ in $\mathcal{L}(F^{\mathcal{P}})$.

A premorphism in $\mathcal{L}(F)^{\mathcal{P}}$ consists of a family

$$F \begin{array}{ccc} & A & \\ x_k \nearrow & & \searrow u_k \\ & \xi_k \Downarrow & \\ y_k \searrow & B & \nearrow v_k \\ & C_k & \end{array}, \quad k \in \mathcal{P}.$$

From lemma 2.3 we can assume all the u_k, v_k, C_k to be equal to a single u, v, C . But this is the data for a premorphism in $\mathcal{L}(F^{\mathcal{P}})$. For the naturality equations we proceed as in the proof of faithfulness in (b).

Here axiom FF1 is inevitable (lemma 2.3), and plays the role of axiom PS2 in the filtered category case. Notice that a function between sets viewed as a functor between trivial categories is injective precisely when it is a full functor. $\square$

We state now an important corollary of theorem 2.4. Let $\mathcal{Cat}_{f\ell}$ be the 2-category of finitely complete categories and finite limit preserving functors. We have:

2.5 Theorem. *Let $\mathcal{A}$ be a 2-filtered 2-category, and $\mathcal{A} \xrightarrow{F} \mathcal{Cat}_{f\ell}$ a 2-functor. Then, the category $\mathcal{L}(F)$ has finite limits, the pseudocone functors $FA \xrightarrow{\lambda_A} \mathcal{L}(F)$ preserve finite limits and induce an equivalence of categories $\mathcal{Cat}_{f\ell}[\mathcal{L}(F), \mathcal{X}] \xrightarrow{\cong} \mathcal{PC}_{f\ell}[F, \mathcal{X}]$; this equivalence is actually an isomorphism.* $\square$

Kennison notion of bifiltered 2-category

In [3], Kennison considers the following notion:

2.6 Definition (Kennison). A 2-category $\mathcal{A}$ is defined to be *bifiltered* when it satisfies the following three axioms:

BF0. Given two objects A, B , there exists C and $A \rightarrow C, B \rightarrow C$.

BF1. Given two arrows $A \xrightarrow[f]{g} B$, there exists $B \xrightarrow{u} C$ and an invertible 2-cell $\gamma : uf \cong ug$.

BF2. Given two 2-cells $A \xrightarrow[\gamma_1 \Downarrow \gamma_2 \Downarrow]{} B$, there exists $B \xrightarrow{u} C$ such that $u\gamma_1 = u\gamma_2$.

This notion of bifiltered 2-category is equivalent to our notion of 2-filtered 2-category. The proof of this is elementary although not entirely trivial, and we leave it as an interesting exercise.

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- [2] M. Artin, A. Grothendieck, J. L. Verdier, *SGA4, 1963/64* Springer Lecture Notes Vol 270 (1972).
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